

Run 03 | NME

Global (including India)

IV only

Target Necessity-motivated entrepreneurship; binary 1 vs. 0

Sample 1,342,884 respondents **Coverage** 89 countries; 2015–2022

Positive cases 69,931 (5.21%) **Case time** 25m 38s

Train / val. / test 913,163 / 161,145 / 268,576 (68% / 12% / 20%, stratified)

Predictors (13): Self-efficacy, opportunity perception, fear of failure, knowing entrepreneurs, intention, media coverage, business exit, business angel, established business owner, good career choice, ease of starting a business, high status, social enterprise.

Controls (0): None. Country and respondent ID are excluded as predictors.

Protocol: Ten CPU models; model seed 20260925; fixed splits by sample/outcome; training-only preprocessing; F1 threshold selected on validation data. Test scores below are unweighted.

Result in brief

Highest observed ROC-AUC: XGBoost (0.9628). Highest F1: XGBoost (0.5395). The all-negative accuracy baseline is 94.79% (full sample).

Model performance on held-out test records

Model	ROC-AUC	PR-AUC	Acc.%	Prec.%	Recall%	F1	Time
XGBoost	0.9628	0.4890	92.56	39.80	83.68	0.5395	16s
CatBoost	0.9627	0.4882	92.52	39.69	83.90	0.5389	1m 14s
Random Forest	0.9626	0.4879	92.52	39.64	83.50	0.5376	1m 25s
H2O GBM	0.9619	0.4761	92.52	39.60	83.28	0.5368	36s
Elastic Net	0.9606	0.4614	92.42	39.21	82.67	0.5319	7m 25s
Lasso	0.9606	0.4610	92.41	39.18	82.71	0.5318	6m 04s
Logistic Regression	0.9605	0.4607	92.42	39.20	82.59	0.5317	54s
Ridge	0.9601	0.4544	92.43	39.20	82.35	0.5311	5m 24s
Decision Tree	0.9536	0.3850	92.24	38.27	80.05	0.5178	6s
Naive Bayes	0.9140	0.3006	89.64	30.02	74.32	0.4277	1m 54s

Models ranked by ROC-AUC; small differences are descriptive, not significance tests. Time includes fit, prediction and output processing. Logistic regression uses a small stabilizing penalty; Lasso/Ridge/Elastic Net use 3-fold training CV.

Random-forest feature importance: top 8

Rank	Predictor	Raw importance	Share%
1	Entrepreneurial intention	17,424.84	56.50
2	Established business owner	6,304.56	20.44
3	Self-efficacy	1,129.55	3.66
4	Knowing entrepreneurs	879.80	2.85
5	Business exit	576.64	1.87
6	Good career choice	413.35	1.34
7	Self-efficacy (missing indicator)	366.69	1.19
8	Ease of starting a business (missing indicator)	317.66	1.03

Share is the percentage of total forest impurity importance across all 25 encoded predictors, not explained variance. Raw values are not comparable across samples. Missing indicators are derived preprocessing features.

Source: Completed Objective 1 batch, 27 September 2026; 03_obj1_global_iv_NME. Scores and importance are taken directly from saved CSV outputs.